

Legendre Polynomials and Functions

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Introduction

Legendre's differential equation arises from separation of variables in spherical coordinates. For example, the azimuthally symmetric angular equation is

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \lambda \Theta = 0.$$

Set $x = \cos \theta$ and write $\Theta(\theta) = y(x)$. The chain rule gives every step in the change of variable:

$$\begin{aligned} \frac{d\Theta}{d\theta} &= \frac{dy}{dx} \frac{dx}{d\theta} = -\sin \theta y', \\ \sin \theta \frac{d\Theta}{d\theta} &= -(1 - x^2)y', \\ \frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) &= -\frac{d}{dx} [-(1 - x^2)y'] \\ &= (1 - x^2)y'' - 2xy'. \end{aligned}$$

Introduction (cont.)

Requiring a nonzero solution that remains bounded at both poles $x = \pm 1$ turns this singular Sturm–Liouville problem into an eigenvalue problem. Its eigenvalues are $\lambda = n(n+1)$, $n = 0, 1, 2, \dots$, and hence

$$(1 - x^2)y'' - 2xy' + n(n+1)y = 0. \quad (\text{C.1.1})$$

On an open interval not containing the singular endpoints, two linearly independent solutions are the Legendre functions of the first and second kinds, denoted by $P_n(x)$ and $Q_n(x)$. For nonnegative integer n , P_n is the degree- n solution that is finite at both endpoints, whereas Q_n is singular there.

The polynomial $P_n(x)$ is defined for every real or complex x ; the interval $[-1, 1]$ is its angular and orthogonality domain, not the domain of the polynomial itself. These ordinary Legendre polynomials express the polar-angle dependence of azimuthally symmetric separated solutions of Laplace's and Helmholtz equations; nonzero azimuthal order uses the associated functions introduced below. Exact spherical symmetry would retain only $n = 0$; general spherical-coordinate problems need the higher angular orders.

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Legendre Polynomials

The first few Legendre polynomials are

$$\begin{aligned}P_0(x) &= 1, \\P_1(x) &= x, \\P_2(x) &= \frac{1}{2}(3x^2 - 1), \\P_3(x) &= \frac{1}{2}(5x^3 - 3x), \\P_4(x) &= \frac{1}{8}(35x^4 - 30x^2 + 3), \\P_5(x) &= \frac{1}{8}(63x^5 - 70x^3 + 15x).\end{aligned}\tag{C.2.1}$$

They follow from Rodrigues' formula

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n, \quad n = 0, 1, 2, \dots \tag{C.2.2}$$

That this formula produces a solution of (C.1.1) is verified directly. With $w = (x^2 - 1)^n$ one has $(1 - x^2)w' + 2nxw = 0$; differentiating $n + 1$ times by the Leibniz rule gives

$$\begin{aligned}(1 - x^2)w^{(n+2)} - 2(n+1)xw^{(n+1)} - n(n+1)w^{(n)} \\ + 2nxw^{(n+1)} + 2n(n+1)w^{(n)} = 0,\end{aligned}$$

Legendre Polynomials (cont.)

which collapses to

$$(1 - x^2)w^{(n+2)} - 2xw^{(n+1)} + n(n+1)w^{(n)} = 0,$$

so $w^{(n)}$, and hence P_n , satisfies Legendre's equation. This licenses the use of (C.2.5) for the Rodrigues polynomials in the orthogonality and spectrum arguments below.

Parity, endpoint values, and recurrence

The generating function is

$$g(x, t) = \frac{1}{\sqrt{1 - 2xt + t^2}} = \sum_{n=0}^{\infty} P_n(x) t^n, \quad (C.2.3)$$
$$|t| < 1, \quad -1 \leq x \leq 1.$$

The identification of the expansion coefficients with the Rodrigues polynomials is proved, not assumed. For $|t|$ small enough that $|2xt - t^2| < 1$, the binomial series gives

$$g(x, t) = \sum_{k \geq 0} \frac{(2k)!}{4^k (k!)^2} t^k (2x - t)^k,$$

and collecting the coefficient of t^n from $k = n - j$, $0 \leq j \leq n/2$, yields

$$\sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(-1)^j (2n - 2j)!}{2^n j! (n - j)! (n - 2j)!} x^{n-2j}.$$

Parity, endpoint values, and recurrence (cont.)

Expanding (C.2.2) through $(x^2 - 1)^n = \sum_j \binom{n}{j} (-1)^{n-j} x^{2j}$ produces the same sum, so the coefficient of t^n is exactly P_n . For fixed $x \in [-1, 1]$ the roots of $1 - 2xt + t^2$ are $t = x \pm i\sqrt{1 - x^2}$, both of modulus 1; hence the Taylor series converges for $|t| < 1$. It makes the parity and endpoint values transparent:

$$\begin{aligned} g(-x, t) &= g(x, -t) \\ &= \sum_{n=0}^{\infty} (-1)^n P_n(x) t^n, \end{aligned} \qquad P_n(-x) = (-1)^n P_n(x),$$

$$g(1, t) = (1 - t)^{-1} = \sum_{n=0}^{\infty} t^n, \qquad P_n(1) = 1,$$

$$\begin{aligned} g(-1, t) &= (1 + t)^{-1} \\ &= \sum_{n=0}^{\infty} (-1)^n t^n, \end{aligned} \qquad P_n(-1) = (-1)^n.$$

Parity, endpoint values, and recurrence (cont.)

In particular, $P_n(0) = 0$ for odd n . Differentiating (C.2.3) with respect to t gives

$$(1 - 2xt + t^2) \frac{\partial g}{\partial t} = (x - t)g.$$

Substitution of $g = \sum_{n \geq 0} P_n t^n$ and comparison of the coefficient of t^n yields

$$(n+1)P_{n+1} - 2nxP_n + (n-1)P_{n-1} = xP_n - P_{n-1},$$

and therefore

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x), \quad n \geq 1. \quad (\text{C.2.4})$$

Orthogonality and normalization

Equation (C.1.1) has the self-adjoint form

$$\frac{d}{dx} [(1-x^2)P'_n(x)] + n(n+1)P_n(x) = 0. \quad (\text{C.2.5})$$

Write this equation once for P_n , once for P_m , multiply by P_m and P_n , respectively, and subtract. The product rule then gives

$$\begin{aligned} \frac{d}{dx} \{ (1-x^2) [P_m P'_n - P_n P'_m] \} \\ + [n(n+1) - m(m+1)] P_m P_n = 0. \end{aligned}$$

After integration from -1 to 1 , the boundary term vanishes because P_n, P_m and their derivatives are finite while $1-x^2=0$ at both endpoints. For $m \neq n$, the eigenvalues are distinct, so

$$\int_{-1}^1 P_m(x) P_n(x) dx = 0.$$

Orthogonality and normalization (cont.)

For $m = n$, Rodrigues' formula supplies the norm. Integrating by parts n times (all boundary terms vanish because $(x^2 - 1)^n$ has zeros of order n at $x = \pm 1$) gives

$$\begin{aligned}\int_{-1}^1 P_n^2 dx &= \frac{(-1)^n}{2^n n!} \int_{-1}^1 P_n^{(n)}(x)(x^2 - 1)^n dx \\ &= \frac{(2n)!}{2^{2n}(n!)^2} \int_{-1}^1 (1 - x^2)^n dx.\end{aligned}$$

Here Rodrigues' formula gives $P_n^{(n)} = (2n)!/(2^n n!)$. The beta integral evaluates to

$$\int_{-1}^1 (1 - x^2)^n dx = \frac{2^{2n+1}(n!)^2}{(2n+1)!},$$

so the complete orthogonality and normalization relation, with δ_{mn} the Kronecker delta, is

$$\int_{-1}^1 P_m(x)P_n(x) dx = \frac{2}{2n+1} \delta_{mn}, \quad m, n = 0, 1, 2, \dots \quad (\text{C.2.6})$$

Orthogonality and normalization (cont.)

Consequently, if $f \in L^2[-1, 1]$, its Legendre-series coefficients are obtained by multiplying the expansion by P_m and using (C.2.6):

$$f(x) \sim \sum_{n=0}^{\infty} a_n P_n(x), \quad a_n = \frac{2n+1}{2} \int_{-1}^1 f(x) P_n(x) dx. \quad (\text{C.2.7})$$

Pointwise convergence requires the usual additional regularity conditions; square-integrability guarantees convergence in the L^2 sense.

Orthogonality and normalization

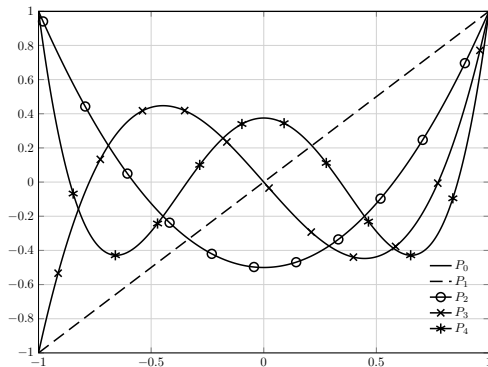


Figure C.1: Legendre polynomials for $n = 0, 1, 2, 3, 4$.

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Associated Legendre Functions

For $-1 \leq x \leq 1$, this book uses the phase-free engineering convention

$$P_n^m(x) = (1 - x^2)^{m/2} \frac{d^m P_n(x)}{dx^m}, \quad 0 \leq m \leq n. \quad (\text{C.3.1})$$

Many mathematical-physics references include the Condon–Shortley factor $(-1)^m$. Thus the convention here and that convention are related by

$$P_{n,\text{here}}^m(x) = (-1)^m P_{n,\text{CS}}^m(x).$$

No differential equation or squared normalization changes under this constant phase. In particular, the convention used here gives the identity employed in Chapter 4,

$$P_n^1(\cos \theta) = \sin \theta P_n'(\cos \theta) = -\frac{d}{d\theta} P_n(\cos \theta).$$

To derive the associated equation without skipping the differentiation step, let $u = d^m P_n / dx^m$. Differentiating (C.1.1) m times and applying the product rule gives

$$(1 - x^2)u'' - 2(m + 1)xu' + [n(n + 1) - m(m + 1)]u = 0. \quad (\text{C.3.2})$$

Associated Legendre Functions (cont.)

Now put $s = 1 - x^2$ and $v = s^{m/2}u = P_n^m$, so that

$$\begin{aligned}u &= s^{-m/2}v, \\u' &= s^{-m/2} \left(v' + \frac{mx}{s}v \right), \\u'' &= s^{-m/2} \left[v'' + \frac{2mx}{s}v' + \frac{m(1 + (m+1)x^2)}{s^2}v \right].\end{aligned}$$

Substitution into (C.3.2) and collection of the v' and v terms gives

$$(1 - x^2)v'' - 2xv' + \left[n(n+1) - \frac{m^2}{1 - x^2} \right] v = 0. \quad (\text{C.3.3})$$

With $x = \cos \theta$, the regular polar factor is $P_n^m(\cos \theta)$. For $m > 0$, multiplication by either real factor $\cos(m\phi)$ or $\sin(m\phi)$, or equivalently use of the complex pair $e^{\pm im\phi}$, supplies the azimuthal dependence. For $m = 0$, the sole nonzero azimuthal factor is the constant 1; $\sin(0\phi)$ is identically zero and is not a second solution.

Associated Legendre Functions (cont.)

For fixed m , equation (C.3.3) has the self-adjoint form

$$\frac{d}{dx} [(1-x^2)v_n'] + \left[n(n+1) - \frac{m^2}{1-x^2} \right] v_n = 0, \quad v_n = P_n^m.$$

Write this equation for v_n and v_l , multiply by v_l and v_n , and subtract:

$$\frac{d}{dx} \{ (1-x^2) [v_l v_n' - v_n v_l'] \} + [n(n+1) - l(l+1)] v_l v_n = 0.$$

After integration on $[-1, 1]$, the boundary term vanishes. For $m = 0$ this is the endpoint argument already used for P_n ; for $m \geq 1$, $v_n = (1-x^2)^{m/2}$ times a polynomial, so each boundary product is $O((1-x^2)^m)$. Thus distinct degrees are orthogonal for fixed m . The norm follows by integrating by parts m times:

$$\begin{aligned} \int_{-1}^1 [P_n^m(x)]^2 dx &= \int_{-1}^1 (1-x^2)^m [P_n^{(m)}(x)]^2 dx \\ &= (-1)^m \int_{-1}^1 P_n(x) \end{aligned}$$

Associated Legendre Functions (cont.)

$$\times \frac{d^m}{dx^m} \left[(1-x^2)^m P_n^{(m)}(x) \right] dx.$$

The required differential identity follows from the self-adjoint form of (C.3.2): with $u = P_n^{(m)}$ that equation reads

$$\frac{d}{dx} \left[(1-x^2)^{m+1} P_n^{(m+1)} \right] = -(n-m)(n+m+1)(1-x^2)^m P_n^{(m)}.$$

Writing $F_m = (1-x^2)^m P_n^{(m)}$, this is $F_m = -F'_{m+1}/[(n-m)(n+m+1)]$; differentiating m times and iterating up to F_n gives

$$F_m^{(m)} = \frac{(-1)^{n-m}(n+m)!}{(n-m)!(2n)!} F_n^{(n)}.$$

Rodrigues' formula makes $P_n^{(n)} = (2n)!/(2^n n!)$ constant, so $F_n = (2n)!/(2^n n!) (1-x^2)^n$ and, by (C.2.2) again, $F_n^{(n)} = (-1)^n (2n)! P_n$. Combining,

$$\frac{d^m}{dx^m} \left[(1-x^2)^m P_n^{(m)}(x) \right] = (-1)^m \frac{(n+m)!}{(n-m)!} P_n(x).$$

Associated Legendre Functions (cont.)

Using (C.2.6), one obtains

$$\int_{-1}^1 P_l^m(x) P_n^m(x) dx = \frac{2}{2n+1} \frac{(n+m)!}{(n-m)!} \delta_{ln}, \quad (\text{C.3.4})$$

for $l, n \geq m \geq 0$. Equivalently, $x = \cos \theta$ and $dx = -\sin \theta d\theta$ give

$$\int_0^\pi [P_n^m(\cos \theta)]^2 \sin \theta d\theta = \frac{2}{2n+1} \frac{(n+m)!}{(n-m)!}. \quad (\text{C.3.5})$$

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Legendre Functions of the Second Kind

On $-1 < x < 1$, the real second solution (more precisely, the Ferrers function of the second kind) is

$$\begin{aligned} Q_n(x) = & \frac{1}{2} P_n(x) \ln \left(\frac{1+x}{1-x} \right) \\ & - \sum_{k=1}^n \frac{1}{k} P_{k-1}(x) P_{n-k}(x), \\ & n = 0, 1, 2, \dots \end{aligned} \tag{C.4.1}$$

where the sum is empty for $n = 0$. For example,

$$\begin{aligned} Q_0(x) &= \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right), \\ Q_1(x) &= \frac{x}{2} \ln \left(\frac{1+x}{1-x} \right) - 1, \\ Q_2(x) &= \frac{3x^2 - 1}{4} \ln \left(\frac{1+x}{1-x} \right) - \frac{3x}{2}. \end{aligned}$$

Legendre Functions of the Second Kind (cont.)

The finite convolution in (C.4.1) is derived as follows rather than verified post hoc. Write $Q_n = \frac{1}{2}P_n L - W_n$ with $L = \ln[(1+x)/(1-x)]$ and W_n a polynomial, and let $\mathcal{L}_n = D[(1-x^2)D] + n(n+1)$. Since $(1-x^2)L' = 2$ is constant, the product rule gives

$$\begin{aligned}\mathcal{L}_n\left[\frac{1}{2}P_n L\right] &= \frac{1}{2}L\mathcal{L}_n[P_n] + (1-x^2)L'P'_n \\ &\quad + \frac{1}{2}P_n[(1-x^2)L']' = 2P'_n,\end{aligned}$$

so Q_n solves (C.1.1) exactly when $\mathcal{L}_n[W_n] = 2P'_n$. For the convolution $W_n = \sum_{k=1}^n \frac{1}{k}P_{k-1}P_{n-k}$, applying the product rule and the Legendre equation to each factor gives

$$\mathcal{L}_n[P_{k-1}P_{n-k}] = 2k(n-k+1)P_{k-1}P_{n-k} + 2(1-x^2)P'_{k-1}P'_{n-k},$$

so the required identity is

$$\sum_{k=1}^n \left[(n-k+1)P_{k-1}P_{n-k} + \frac{1-x^2}{k}P'_{k-1}P'_{n-k} \right] = P'_n. \quad (\text{C.4.2})$$

Legendre Functions of the Second Kind (cont.)

Multiplication by t^n and summation over $n \geq 1$ converts this to a generating-function identity. With $g = \sum_{n \geq 0} P_n t^n$ and $H = \sum_{n \geq 0} P_n t^{n+1}/(n+1)$, so that $H_t = g$ and $H(x, 0) = 0$, the left side of (C.4.2) sums to $tg(tg)' + (1-x^2)H_x g_x$. From $g = (1-2xt+t^2)^{-1/2}$, direct differentiation gives $(tg)' = (1-xt)g^3$ and $g_x = tg^3$; and $\partial_t[(1-x^2)H_x] = (1-x^2)g_x = \partial_t[1 - (1-xt)g]$ with both sides vanishing at $t = 0$, hence $(1-x^2)H_x = 1 - (1-xt)g$. Therefore

$$\begin{aligned}tg(tg)' + (1-x^2)H_x g_x &= t(1-xt)g^4 + tg^3 - t(1-xt)g^4 \\&= tg^3 = g_x,\end{aligned}$$

and the coefficient of t^n in $g_x = \sum_{n \geq 0} P'_n t^n$ is P'_n , proving (C.4.2). Hence Q_n solves (C.1.1); alternatively, these functions follow from Q_0, Q_1 using the same three-term recurrence (C.2.4). To retain the Wronskian sign, divide the differential equation by $1-x^2$ and set $W = P_n Q'_n - P'_n Q_n$. Abel's identity gives

$$W' = \frac{2x}{1-x^2} W \quad \implies \quad W = \frac{C}{1-x^2}.$$

Legendre Functions of the Second Kind (cont.)

As $x \rightarrow 1^-$, $P_n \rightarrow 1$ and the leading logarithmic term in Q_n gives $(1 - x^2)Q'_n \rightarrow 1$, whereas $(1 - x^2)P'_n Q_n \rightarrow 0$; hence $C = 1$. Therefore

$$P_n Q'_n - Q_n P'_n = \frac{1}{1 - x^2},$$

so P_n and Q_n are linearly independent on $(-1, 1)$. The logarithm makes Q_n singular at $x = \pm 1$.

The real Legendre function on either exterior interval $|x| > 1$ uses a different real logarithmic expression,

$$\begin{aligned} Q_n(x) &= \frac{1}{2} P_n(x) \ln \left(\frac{x+1}{x-1} \right) \\ &\quad - \sum_{k=1}^n \frac{1}{k} P_{k-1}(x) P_{n-k}(x), \quad |x| > 1. \end{aligned} \tag{C.4.3}$$

for which the logarithm's ratio is positive. Complex analytic continuation requires a specified branch and must not be obtained by naively inserting $|x| > 1$ into the interior principal-log formula. The exterior $Q_n(\xi)$, with ξ the spheroidal radial coordinate, occurs, for example, as a radial factor in prolate-spheroidal separations. For a spherical exterior problem, by contrast, the radial Laplace factors are $r^{-(n+1)}$, not $Q_n(\cos \theta)$.

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Summary of Important Relations

For $n = 0, 1, 2, \dots$, and $0 \leq m \leq n$, the principal relations are

$$\begin{aligned}\frac{1}{\sqrt{1-2xt+t^2}} &= \sum_{n=0}^{\infty} P_n(x)t^n, \\ |t| &< 1, \quad -1 \leq x \leq 1, \\ \int_{-1}^1 P_m(x)P_n(x) dx &= \frac{2}{2n+1} \delta_{mn}, \\ \frac{d}{dx} [(1-x^2)P'_n(x)] + n(n+1)P_n(x) &= 0, \\ \int_0^\pi [P_n^m(\cos \theta)]^2 \sin \theta d\theta \\ &= \frac{2}{2n+1} \frac{(n+m)!}{(n-m)!}.\end{aligned}$$

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Bridge Proofs for Spherical Angular Functions

This section closes the gaps used by Chapter 4: that the regular eigenvalues are exactly $\lambda = n(n+1)$, the admissible associated range $n \geq m$, the angular-Laplacian eigenvalue of the full (polar–azimuthal) mode, and the two angular power identities behind the scattering cross section.

Regular spectrum. Let Θ be nonzero and regular at both poles with $[(1-u^2)\Theta']' + \lambda\Theta = 0$. Subtracting the self-adjoint equation (C.2.5) for P_n cross-multiplied by Θ and P_n and integrating gives

$$(\lambda - n(n+1)) \int_{-1}^1 \Theta P_n du = 0, \quad n = 0, 1, 2, \dots,$$

the boundary term vanishing by the endpoint factor $1-u^2$. Unless $\lambda = n(n+1)$ for some n , every Legendre coefficient of Θ vanishes, and polynomial density (the Weierstrass theorem, taken as an analysis prerequisite) with the orthogonality (C.2.6) forces $\Theta \equiv 0$. Hence the regular spectrum is exactly $\{n(n+1)\}$. This supplies the necessity argument behind series termination in Chapter 4.

Admissible associated range. Writing the associated trial factor as $(1-u^2)^{m/2}V(u)$ with $V(u) = \sum_q b_q u^q$ gives $(1-u^2)V'' - 2(m+1)uV' + [\lambda - m(m+1)]V = 0$, whose

Bridge Proofs for Spherical Angular Functions (cont.)

coefficient recurrence carries the numerator $(q + m)(q + m + 1) - \lambda$. A regular polynomial V therefore requires $\lambda = (N + m)(N + m + 1)$ for the terminating degree N , i.e.

$$n = N + m \geq m.$$

That termination constructs the candidates; that no further regular eigenfunctions exist follows the regular-spectrum argument. For fixed m , cross-multiplying the associated equation for a regular eigenfunction Θ with that for P_n^m and integrating gives

$$(\lambda - n(n + 1)) \int_{-1}^1 \Theta P_n^m du = 0, \quad n = m, m + 1, \dots,$$

the boundary term $(1 - u^2)(\Theta P_n^{m'} - P_n^m \Theta')$ vanishing: for $m = 0$ the endpoint factor $1 - u^2$ multiplies bounded derivatives, and for $m \geq 1$ regularity of $\Theta = (1 - u^2)^{m/2} V$ gives $(1 - u^2)\Theta' = O((1 - u^2)^{m/2}) \rightarrow 0$. Since the derivatives $\{P_n^{(m)}\}_{n \geq m}$ run through a triangular basis of all polynomials, the regular factor $(1 - u^2)^{m/2}\Theta$ is orthogonal to every polynomial, and polynomial density forces $\Theta \equiv 0$ unless $\lambda = n(n + 1)$ for some $n \geq m$. At such an eigenvalue the polynomial V terminates and gives P_n^m ; the second solution is

Bridge Proofs for Spherical Angular Functions (cont.)

singular at a pole. So for each $m \geq 0$ the regular eigenfunctions are exactly the P_n^m with $n \geq m$, consistent with the convention (C.3.1).

Angular-Laplacian eigenvalue of the full mode. With $\nabla_a^2 = \partial_\theta^2 + \cot \theta \partial_\theta + (\sin \theta)^{-2} \partial_\phi^2$ and $\Theta(\theta) = P_n^m(\cos \theta)$, the associated equation in θ reads $(\sin \theta)^{-1} (\sin \theta \Theta')' + [n(n+1) - m^2 / \sin^2 \theta] \Theta = 0$. Acting on the full mode $\Theta(\theta)e^{im\phi}$ (and hence on its real cosine/sine parts),

$$\nabla_a^2 [P_n^m(\cos \theta)e^{im\phi}] = -n(n+1) P_n^m(\cos \theta)e^{im\phi}. \quad (\text{C.6.1})$$

The azimuthal factor supplies the cancelling $-m^2 / \sin^2 \theta$ term. In particular, the polar factor alone is *not* an eigenfunction:

$$\nabla_a^2 P_n^1(\cos \theta) = \left[-n(n+1) + \frac{1}{\sin^2 \theta} \right] P_n^1(\cos \theta).$$

Only the full $P_n^1(\cos \theta) \cos \phi$ (or sine) mode carries the eigenvalue $-n(n+1)$ cancelled in the Chapter 4 interface matching.

Scalar Spherical-Harmonic Completeness

A scalar angular expansion is needed before discussing the vector basis. With the phase-free associated-Legendre convention of these notes, define for $-n \leq m \leq n$

$$Y_{nm}(\theta, \phi) = \left[\frac{2n+1}{4\pi} \frac{(n-|m|)!}{(n+|m|)!} \right]^{1/2} P_n^{|m|}(\cos \theta) e^{im\phi}.$$

The associated-Legendre norm and Fourier orthogonality give $\int Y_{nm}^* Y_{pq} d\Omega = \delta_{np} \delta_{mq}$. To establish completeness rather than only orthogonality, suppose $f \in L^2(S^2)$ is orthogonal to every Y_{nm} . Put $u = \cos \theta$ and take the azimuthal Fourier coefficient

$$f_m(u) = \frac{1}{2\pi} \int_0^{2\pi} f(u, \phi) e^{-im\phi} d\phi.$$

For each integer m , orthogonality gives $\int_{-1}^1 f_m(u) P_n^{|m|}(u) du = 0$ for all $n \geq |m|$. Since $P_n^{|m|} = (1-u^2)^{|m|/2} P_n^{(|m|)}$, and $\{P_n^{(|m|)} : n \geq |m|\}$ is a triangular basis of all polynomials, the function $g_m = (1-u^2)^{|m|/2} f_m$ is orthogonal to every polynomial. Polynomial density gives $g_m = 0$ almost everywhere. Its prefactor is nonzero on $(-1, 1)$, so $f_m = 0$ almost everywhere. Fourier completeness then gives $f = 0$. Hence the Y_{nm} form a complete orthonormal basis, not merely an orthogonal family.

Scalar Spherical-Harmonic Completeness (cont.)

The smooth-field uses of this expansion also require differentiation. For smooth f , integration by parts with the angular Laplacian gives, for $n \geq 1$ and every positive integer p ,

$$|\langle Y_{nm}, f \rangle| \leq [n(n+1)]^{-p} \|(\nabla_a^2)^p f\|_{L^2}.$$

The required polynomial bounds on basis derivatives follow without an asymptotic formula. Rotations preserve the $(2n+1)$ -dimensional eigenspace, so the basis-independent sum $\sum_m |Y_{nm}(\Omega)|^2$ is constant. Integrating it over the sphere fixes that constant as $(2n+1)/(4\pi)$. For the rotation derivatives $D_j = (\hat{e}_j \times \hat{r}) \cdot \vec{\nabla}_a$, integration by parts and $\sum_j |D_j f|^2 = |\vec{\nabla}_a f|^2$ give

$$\sum_j \|D_j f\|_{L^2}^2 = n(n+1) \|f\|_{L^2}^2$$

within this eigenspace. Each D_j preserves it. Repetition and the pointwise bound from the constant sum therefore bound any fixed number q of rotation derivatives by a constant times $\sqrt{2n+1} [n(n+1)]^{q/2} \|f\|_{L^2}$. On regular coordinate patches these derivatives span the tangential derivatives. Combined with the arbitrarily large p above,

Scalar Spherical-Harmonic Completeness (cont.)

these estimates give uniform convergence of each fixed differentiated expansion of a smooth field.

Angular power identities. Put $U_n(\theta) = P_n^1(\cos \theta)$, $\pi_n = U_n / \sin \theta$, $\tau_n = U'_n$. From (C.3.4) with $m = 1$,

$$\int_0^\pi U_n U_\ell \sin \theta \, d\theta = \frac{2n(n+1)}{2n+1} \delta_{n\ell}.$$

Since $U_n = O(\sin \theta)$ at the poles, integration by parts with the $m = 1$ angular equation gives the diagonal identity. Explicitly, that equation reads $(\sin \theta U'_\ell)' = -[\ell(\ell+1) - \sin^{-2} \theta] U_\ell \sin \theta$, so the derivative half of the integral becomes

$$\begin{aligned} \int_0^\pi U'_n U'_\ell \sin \theta \, d\theta &= [\sin \theta U_n U'_\ell]_0^\pi - \int_0^\pi U_n (\sin \theta U'_\ell)' \, d\theta \\ &= \ell(\ell+1) \int_0^\pi U_n U_\ell \sin \theta \, d\theta - \int_0^\pi \frac{U_n U_\ell}{\sin \theta} \, d\theta, \end{aligned}$$

Scalar Spherical-Harmonic Completeness (cont.)

the boundary term vanishing by $U_n = O(\sin \theta)$. Adding the $\pi_n \pi_\ell$ part cancels the last integral, leaving

$$\begin{aligned} \int_0^\pi (\pi_n \pi_\ell + \tau_n \tau_\ell) \sin \theta \, d\theta &= \ell(\ell + 1) \int_0^\pi U_n U_\ell \sin \theta \, d\theta \\ &= \frac{2n^2(n+1)^2}{2n+1} \delta_{n\ell}, \end{aligned} \tag{C.6.2}$$

while the mixed combination needs no spectrum at all:

$$\int_0^\pi (\pi_n \tau_\ell + \tau_n \pi_\ell) \sin \theta \, d\theta = [U_n U_\ell]_0^\pi = 0. \tag{C.6.3}$$

With the azimuthal integrals $\int_0^{2\pi} \cos^2 \phi \, d\phi = \int_0^{2\pi} \sin^2 \phi \, d\phi = \pi$, the diagonal weight against the $[(2n+1)/(n(n+1))]^2$ amplitude prefactors reduces to $2(2n+1)$, yielding the modal scattering sum used in Chapter 4 with all cross terms eliminated.

References



George B. Arfken, Hans J. Weber, and Frank E. Harris.
Mathematical Methods for Physicists: A Comprehensive Guide.
Academic Press, 7th edition, 2011.
Electronic edition.



Constantine A. Balanis.
Advanced Engineering Electromagnetics.
Wiley, 2nd edition, 2012.



Kasra Barkeshli.
Advanced Electromagnetics and Scattering Theory.
Springer, 1st edition, 2015.
Edited by Sina Khorasani.



Craig F. Bohren and Donald R. Huffman.
Absorption and Scattering of Light by Small Particles.
Wiley Science Series. John Wiley & Sons, 2008.
Revised electronic reprint of the 1983 original.

References (cont.)



Weng Cho Chew.

Waves and Fields in Inhomogeneous Media.

IEEE Press Series on Electromagnetic Wave Theory. Wiley-IEEE Press, 1st edition, 1999.

IEEE Press reprint of the 1990 version with corrections and minor updating.



Phillip C. Clemmow.

The Plane Wave Spectrum Representation of Electromagnetic Fields, volume 12 of *International Series of Monographs in Electromagnetic Waves*.

Elsevier Science, 2013.

Electronic reissue of the 1966 Pergamon Press first edition.



Robert E. Collin.

Field Theory of Guided Waves.

IEEE Press, 2nd edition, 1990.

References (cont.)



Georges A. Deschamps.

Electromagnetics and differential forms.

Proceedings of the IEEE, 69(6):676–696, June 1981.



NIST Digital Library of Mathematical Functions.

<https://dlmf.nist.gov/>, Release 1.2.7 of 2026-06-15.

F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, eds.



Stuart Gregson, John McCormick, and Clive Parini.

Principles of Planar Near-Field Antenna Measurements.

Number 53 in IET Electromagnetic Waves Series. Institution of Engineering and Technology, 2007.



Roger F. Harrington.

Time-Harmonic Electromagnetic Fields.

IEEE Press Series on Electromagnetic Wave Theory. IEEE Press, 2001.

Classic reissue of the 1961 original.

References (cont.)



Akira Ishimaru.

Electromagnetic Wave Propagation, Radiation, and Scattering: From Fundamentals to Applications.

Wiley-IEEE Press, 2nd edition, 2017.



John David Jackson.

Classical Electrodynamics.

Wiley India, 3rd edition, 2012.

Indian reprint of the third edition.



Shyh-Kang Jeng.

Scattering from a cavity-backed slit in a ground plane—TE case.

IEEE Transactions on Antennas and Propagation, 38(10):1523–1529, October 1990.



Jian-Ming Jin.

Theory and Computation of Electromagnetic Fields.

Wiley-IEEE Press, 2nd edition, 2015.

References (cont.)



Milton Kerker.

The Scattering of Light and Other Electromagnetic Radiation, volume 16 of *Physical Chemistry: A Series of Monographs*.

Academic Press, 2013.

Series edited by Ernest M. Loebl; electronic reissue of the 1969 first edition.



Jin Au Kong.

Theory of Electromagnetic Waves.

Wiley-Interscience, New York, 1975.



Jin Au Kong.

Electromagnetic Wave Theory.

EMW Publishing, Cambridge, MA, 2008.



Ismo V. Lindell.

Differential Forms in Electromagnetics.

Wiley-IEEE Press, 2004.

References (cont.)



Kyle A. Novak and Laura J. Fox.

Special Functions of Mathematical Physics: A Tourist's Guidebook.

Equal Share Press, LLC, 2019.



Lukas Novotny and Bert Hecht.

Principles of Nano-Optics.

Cambridge University Press, 2nd edition, 2012.



Andrey V. Osipov and Sergei A. Tretyakov.

Modern Electromagnetic Scattering Theory with Applications.

John Wiley & Sons, 1st edition, 2017.



Prabhakar H. Pathak and Robert J. Burkholder.

Electromagnetic Radiation, Scattering, and Diffraction.

Wiley-IEEE Press, 2021.

References (cont.)



Xin-Qing Sheng.

Electromagnetic Theory, Calculations, and Applications.

Tsinghua University Press, Beijing, 2nd edition, 2023.

In Chinese.



Xin-Qing Sheng and Wei Song.

Essentials of Computational Electromagnetics.

Wiley, 2012.



Glenn S. Smith.

An Introduction to Classical Electromagnetic Radiation.

Cambridge University Press, 1st edition, 1997.



Carlo G. Someda.

Electromagnetic Waves.

CRC Press, 2nd edition, 2006.

References (cont.)



Chen-To Tai.

General Vector and Dyadic Analysis: Applied Mathematics in Field Theory.
Wiley-IEEE Press, 2nd edition, 1997.



H. C. van de Hulst.

Light Scattering by Small Particles.
Dover Publications, 1981.
Reprint of the 1957 Wiley edition.



Pochi Yeh.

Optical Waves in Layered Media.
Wiley Series in Pure and Applied Optics. Wiley, 2005.
Paperback reissue of the 1988 original.



Andrew Zangwill.

Modern Electrodynamics.
Cambridge University Press, 2013.